COC/POE LEO Encapsulant Lifetime Simulator — Model Summary

Team Monomers · Engineering Design · LEO (ISS 400km, 51.6°) · 15-year mission · 7 channels integrated at dt=0.1y

Inputs & Environment

Input	Range	Default	Env (LEO)	Value	Source
NB (norbornene mol%)	20–55	40	Rad rate	0.5 kGy/y	MDPI 12(19)4058
TH (total thickness µm)	100–1000	400	UV rate	5256 ESH/y	NREL Solar Spectra
rPOE (POE fraction %)	20–80	50	AO fluence	3.28e21 atoms/cm^2/y	NASA NTRS 20070002707
rCOC (auto)	100−rPOE		ΔT / cycles	300°C / 5840 per y	PCBSync

Equations (7 channels)

Ch1 AO erosion: dh_gross=Ey · F_AO · 1e4 · χ; POE first, then COC × μ_core; fail h_COC ≤ 0
Ch2 Roughness: L_rough=k_rough · Δh_tot (couples into Ch3)
Ch3 Optical: T(t)=T0 · exp(-α · TH) · exp(-k_rad · rad · t · 100) · exp(-k_UV · UV · t) · exp(-L_rough) · exp(-k_cont · CVCM_eff); fail T<T0 · 0.80
Ch4 Modulus: E_base=E_COC · f_COC+E_POE · f_POE; E(t)=E_base · (1+k_xl · rad · t · 100)
Ch5 Static σ: σ=E(t) · |α_POE−α_COC| · 1e-6 · ΔT; fail σ>σ_adh (σ_adh=100 MPa)
Ch6 Fatigue: D(t)=(N_cyc · t/N_ref) · (σ/σ_adh)^n; fail D ≥ 1; N_ref=2e4, n=2
Ch7 Tg softening: Tg=Tg0+m_g · NB (Tg0=−15, m_g=4.5); fail Tg<150°C → NB ≥ 36.7
Ch8 Outgassing: TML_sat · (1−e^(−t/τ)) ≥ 1.0% OR CVCM_sat · (1−e^(−t/τ)) · (1−γ_AO) ≥ 0.1%

Key Material Constants (composition-weighted)

Param	POE	COC	Param	POE	COC
Young E (MPa)	10	3000	TML sat (%)	1.5	0.3
CTE (ppm/°C)	180	60	CVCM sat (%)	0.12	0.03
AO yield Ey	3.5e-24 cm^3/atom (polyolefin, NASA NTRS 20090034484)		τ outgas	0.5 y	

Verification Results (σ_adh = 100 MPa)

Case	NB / TH / rPOE	Life	Dominant Channel
optimal	40 / 400 / 50	3.9 y	Optical (T<80%)
danger	20 / 400 / 50	0 y	Tg softening (instant)
thin	40 / 200 / 50	4.4 y	Optical
thick	40 / 800 / 50	2.7 y	Optical
cocHeavy	40 / 400 / 20	4.1 y	Optical
poeHeavy	40 / 400 / 80	0.7 y	Outgassing TML

Modeling Notes & Assumptions

- The constraint-relaxation factor η was dropped (insufficient physical basis); σ_adh raised to 100 MPa = literature upper bound for surface-treated polyolefin interfacial shear strength. Restores dimensional consistency with σ_full.
- Radiation–UV synergy term k_syn set to 0 (no validated quantitative data).
- Optical channel combines all losses multiplicatively → conservative at large TH; some losses self-attenuate with depth in reality.
- N_ref = 2×10^4 is a conservative undergraduate estimate; to be replaced with experimental polyolefin/glass interface fatigue data.
- Lifetime defined as (t−dt) at first channel threshold crossing; Ch7 (Tg) is instantaneous.

Design Insights

- NB ≥ 37 mol% — required for Tg ≥ 150°C (avoid Ch7).
- rPOE ≤ ~65% — keep TML_sat below 1% ASTM E595 (avoid Ch8).
- Optical dominates the middle region; thinner stacks reduce Beer-Lambert loss.
- rPOE ≈ 50% provides best balance between Ch5 (static σ), Ch6 (fatigue), Ch1 (AO).

Sources: MDPI Electronics 12(19)4058 · NREL Solar Spectra · NASA NTRS 20070002707, 20090034484 · NASA GSFC ASTM E595 · TOPAS Advanced Polymers · ScienceDirect S0376738805006071 · Wiley pip.3323, pip.3551, Polym. Eng. Sci. (1969). Team Monomers, 2026-05-14. GitHub: junjunny/Monomers_COC.